

Lecture 9

Microeconomic Theory 3 – ECON 709

Fall 2026

Recap' lecture 8

- ▶ $\succsim$ is monotone if and only if u is strictly increasing.
- ▶ $\succsim$ is strongly monotone if and only if u is strongly increasing.
- ▶ The preference relation $\succsim$ is **locally non-satiated** if for every $x \in X$ and every $\epsilon > 0$, there exists $y \in X$ such that $\|y - x\| \leq \epsilon$ and $y \succ x$.
- ▶ Three equivalent definitions of convex preferences:
 - ▶ The preference relation $\succsim$ satisfies convexity 1, if $x \succsim y$ implies $\alpha x + (1 - \alpha)y \succsim y$ for all $\alpha \in [0, 1]$.
 - ▶ The preference relation $\succsim$ satisfies convexity 2, if for all x, y and $z = \alpha x + (1 - \alpha)y$ with $\alpha \in [0, 1]$, $z \succsim x$ or $z \succsim y$.
 - ▶ The preference relation $\succsim$ satisfies convexity 3, if for all $x \in X$, the upper contour set $\{y | y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1 - \alpha)z \succsim x$ for any $\alpha \in [0, 1]$.
- ▶ Strict convexity: adjust the three convexity definitions with $\alpha \in (0, 1)$ and a strict preference implication.

Utility Representations of Convex Preferences

- ▶ A function $u : X \rightarrow \mathbb{R}$ is concave if for all x, y , and $\lambda \in [0, 1]$ we have $u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y)$, and is quasi-concave if for all y , the set $\{x | u(x) \geq u(y)\}$ is convex, or equivalently, if $u(\lambda x + (1 - \lambda)y) \geq \min \{u(x), u(y)\}$ for all x, y and $\lambda \in [0, 1]$.
- ▶ Trivially, any concave function is also quasi-concave.

Proposition 13

If a preference relation $\succsim$ is represented by a utility function, then $\succsim$ is convex if and only if u is quasi-concave.

Proof left as exercise.

Thus, a concave utility function implies convex preferences.

However, convexity of a preference relation does not imply that its utility representation is concave!

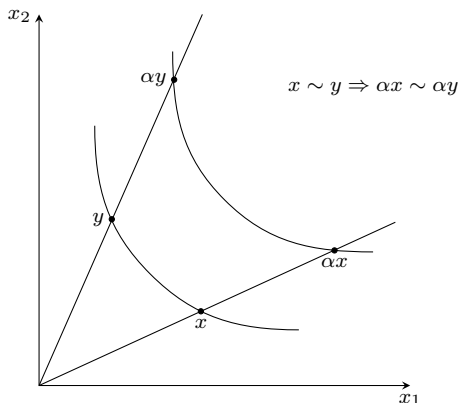
Special Classes of Consumer Preferences

For some classes of preferences, it is possible to deduce the entire preference relation from a single indifference set, for example, homothetic or quasi-linear preferences...

Homothetic Preferences

Definition 18

A monotone preference relation $\succsim$ is **homothetic** if for all x, y , $x \sim y$ implies that $\alpha x \sim \alpha y$ for any $\alpha \geq 0$.



Utility Representation of Homothetic Preferences

- ▶ For example, preferences represented by a “Cobb-Douglas” utility function $\prod_{k=1,\dots,K} x_k^{\beta_k}$ with $\beta_k > 0$ are homothetic.
- ▶ More generally, any preference relation represented by a utility function u that is homogeneous of any degree λ (that is $u(\alpha x) = \alpha^\lambda u(x)$) is homothetic:

$$\begin{aligned} x \sim y &\Leftrightarrow u(x) = u(y) \Leftrightarrow \alpha^\lambda u(x) = \alpha^\lambda u(y) \\ &\Leftrightarrow u(\alpha x) = u(\alpha y) \Leftrightarrow \alpha x \sim \alpha y. \end{aligned}$$

Proposition 14

A monotone preference relation $\succsim$ on $X = \mathbb{R}_+^K$ is continuous and homothetic if and only if $\succsim$ can be represented by a continuous utility function that is homogeneous of degree one.

Proof.

$\Rightarrow$: We already proved for Prop. 8 that for any bundle x there exists a unique bundle $(t(x), \dots, t(x))$ on the main diagonal, so that $x \sim (t(x), \dots, t(x))$, and that the function $u(x) = t(x)$ is a continuous utility representation of $\succsim$. By the assumption that the preferences are homothetic, $\alpha x \sim (\alpha t(x), \dots, \alpha t(x))$ and thus $u(\alpha x) = \alpha t(x) = \alpha u(x)$, that is u is homogeneous of degree 1.

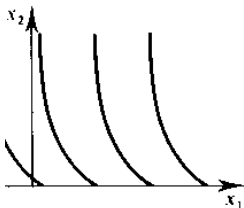
$\Leftarrow$: By our previous results, since u is continuous and homogenous of degree λ , $\succsim$ is continuous and homothetic. □

Quasi-linear Preferences

Definition 19

A preference relation $\succsim$ is **quasi-linear** in commodity 1 (referred to as "numeraire"), if

1. All the indifference sets are parallel displacements of each other along the axis of commodity 1. That is, if $x \sim y$, then $(x + \alpha e_1) \sim (y + \alpha e_1)$ where $e_1 = (1, 0, \dots, 0)$ and any $\alpha \in \mathbb{R}$.
 2. Good 1 is desirable; that is $x + \alpha e_1 \succ x$ for all x and $\alpha > 0$.
- If I is an indifference curve, then the set $I_\alpha = \{x \mid \text{there exists } y \in I \text{ such that } x = y + (\alpha, 0, \dots, 0)\}$ is an indifference curve as well.



The numeraire as “money”

- ▶ The most common interpretation of commodity 1 is “money”, left over to buy other commodities not included in the commodities $2, \dots, K$.
- ▶ Part 1 of the definition implies that adding an equal amount of money to two indifferent bundles does not change the trade off in terms of preferences between the commodities (indifference remains); in other words, there are no “wealth effects”.
- ▶ Part 2 of the definition implies that money is desirable.
- ▶ Quasi-linearity in commodity 1, continuity and monotonicity of $\succsim$ implies that for every $(x_2, \dots, x_K)$ there is a number $v(x_2, \dots, x_K)$ such that $(0, x_2, \dots, x_K) \sim (v(x_2, \dots, x_K), 0, \dots, 0)$. (For a proof refer to Rubinstein.)

That is, for every commodity bundle x with zero money, there exists an amount of money m such that there is indifference between having x and having m .

Utility Representation of Quasi-Linear Preferences

- ▶ If a preference relation $\succsim$ is represented by a utility function $u(x) = x_1 + v(x_2, \dots, x_K)$ for some function $v : \mathbb{R}_+^{K-1} \rightarrow \mathbb{R}$, then $\succsim$ is quasi-linear in commodity 1:
 - ▶ First, $x \sim y \Leftrightarrow u(x) = u(y) \Leftrightarrow u(x) + \alpha = u(y) + \alpha \Leftrightarrow (x_1 + \alpha) + v(x_2, \dots, x_K) = (y_1 + \alpha) + v(y_2, \dots, y_K) \Leftrightarrow u(x + \alpha e_1) = u(y + \alpha e_1) \Leftrightarrow x + \alpha e_1 \sim y + \alpha e_1$.
 - ▶ Second, consider $x + \alpha e_1$ and x where $\alpha > 0$. Then $u(x + \alpha e_1) = x_1 + \alpha + v(x_2, \dots, x_K) > u(x) = x_1 + v(x_2, \dots, x_K)$. Thus $x + \alpha e_1 \succ x$.
- ▶ Intuition for the utility function $u(x) = x_1 + v(x_2, \dots, x_K)$:
 - ▶ The utility from bundle x is some “sub-utility” from $(x_2, \dots, x_K)$ plus the amount of money left over.
 - ▶ If $x \succ y$, then $u(x) - u(y) = m \Leftrightarrow u(x) = m + u(y)$ where m can be interpreted as the amount of money to be added to y to establish an indifference with x .

- ▶ If a preference relation $\succsim$ is monotonic, continuous and quasi-linear in commodity 1, then it can be represented by a utility function of the form $x_1 + v(x_2, \dots, x_K)$:
 - ▶ Remember that for every $(x_2, \dots, x_K)$ there is a number $v(x_2, \dots, x_K)$ such that $(0, x_2, \dots, x_K) \sim (v(x_2, \dots, x_K), 0, \dots, 0)$.
 - ▶ By the quasi-linearity in commodity 1, $(x_1, x_2, \dots, x_K) \sim (x_1 + v(x_2, \dots, x_K), 0, \dots, 0)$.
 - ▶ Thus by desirability of commodity 1, the function $x_1 + v(x_2, \dots, x_K)$ represents $\succsim$.
- ▶ Thus, any continuous and monotonic preference relation that is quasi-linear in the first commodity is consistent with the following procedure: the consumer considers the value (in terms of the first commodity) of a combination of goods 2, ..., K , which is independent of the quantity of the first commodity.
 E.g. the consumer always values two apples and three potatoes at 5 dollars, independent of whether he has 100 or 1000 dollars in his pocket.

Having discussed the consumer's preferences over X , we now move to analyze the consumer choice.

Demand: Consumer Choice

based on: Rubinstein, lecture 5; MWG, sections 2.D, 3.D

Consumer Choice Problem

- ▶ we again assume that the consumer makes a "rational choice": the consumer chooses a "best" bundle out of the bundles which are "affordable" and "available" to her personally (her choice problem), "best" according to her preference relation over X
- ▶ in addition to potential physical constraints captured in X (remember that we assume $X = \mathbb{R}_+^K$), the consumer faces an economic constraint: she can only choose bundles which she can afford
 - ▶ commodities are traded at prices: price vector $p = (p_1, \dots, p_K)$ where p_k is the price for commodity k ; assume $p_k > 0$ for all k ; consumer is assumed to be a *price taker*
 - ▶ consumer has wealth $w > 0$
 - ▶ p and w determine a consumer's choice problem, her *budget set*:

$$B_{p,w} = \{x \in \mathbb{R}_+^K \mid px \leq w\}$$

that is, the set of all commodity bundles which cost weakly less than her wealth w

Budget Set: $B_{p,w} = \{x \in \mathbb{R}_+^K | px \leq w\}$

- ▶ a budget set is compact:
closed since it is defined by weak inequalities and bounded since for any $x \in B_{p,w}$ and for all k , $0 \leq x_k \leq w/p_k$
- ▶ a budget set is convex:
If $x, y \in B_{p,w}$, then $px \leq w$, $py \leq w$, $x_k \geq 0$, and $y_k \geq 0$ for all k . Thus, for all $\alpha \in [0, 1]$,
 $p[\alpha x + (1 - \alpha)y] = \alpha px + (1 - \alpha)py \leq w$ and
 $\alpha x_k + (1 - \alpha)y_k \geq 0$ for all k , that is, $\alpha x + (1 - \alpha)y \in B_{p,w}$.
- ▶ the upper boundary of the budget set is the *budget hyperplane* (budget line, if $K = 2$) given by the set

$$\{x \in \mathbb{R}_+^K | px = w\}$$

- ▶ the budget hyperplane reflects the relative terms of exchange between commodities on the market (e.g. for $K = 2$, the slope of the budget line is $-p_1/p_2$)

Formalization of the consumer problem

CP1: Find $x^* \in B_{p,w}$ such that $x^* \succsim x$ for all $x \in B_{p,w}$.

CP2: If a consumer's preference relation $\succsim$ on X is represented by a utility function u , the consumer's decision problem can be stated as a utility maximization problem (UMP) under constraints:

$$\max_{x \in X} u(x) \text{ subject to } px \leq w$$

- If $\succsim$ is represented by u , CP1 is equivalent to CP2:
 $x^* \succsim x$ for all $x \in B_{p,w} \Leftrightarrow u(x^*) \geq u(x)$ for all $x \in B_{p,w}$.
Bundle x^* is a solution to CP1 iff x^* is a solution to CP2.

Solution to the Consumer Choice Problem

Proposition 15

If $\succsim$ is continuous, then all consumer choice problems have a solution.

Proof.

If $\succsim$ is continuous, then it can be represented by a continuous utility function u . Thus we can state the consumer choice problem as a UMP. Since a continuous function has a maximum value on any compact set, we know that the UMP has a solution. Thus there exists a $\succsim$ -best bundle in $B_{p,w}$ and the consumer choice problem has a solution. □

We denote by $x(p, w)$ the rule that assigns the solution of the consumer choice problem to each price wealth situation (p, w) . That is, given p and w , $x(p, w)$ is the set of $\succsim$ -best bundles in $B_{p,w}$ or

$$x(p, w) = \arg \max_{x \in B_{p,w}} u(x).$$

Unique or multiple solutions to the consumer problem

Depending on the preferences and $B_{p,w}$, there can be a unique $\succsim$ -best bundle ($x(p, w)$ is a singleton given (p, w)) or multiple $\succsim$ -best bundles in $B_{p,w}$.

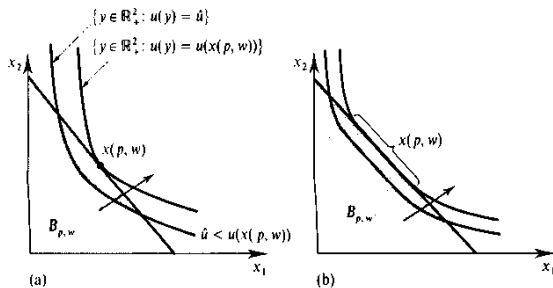


Figure 3.D.1
The utility maximization problem (UMP).
(a) Single solution.
(b) Multiple solutions.

Walrasian Demand Correspondence\Function

- ▶ the rule $x(p, w)$ that assigns the set of optimal consumption bundles to each (p, w) situation is known as "Walrasian/market demand correspondence"
- ▶ when $x(p, w)$ is single-valued for all (p, w) (unique solution to all consumer choice problems), we refer to it as the (Walrasian/market) demand function
- ▶ We assume in the following that $\succsim$ is continuous such that $\succsim$ can always be represented by a continuous utility function. This implies that we can always state the consumer choice problem as a UMP and $x(p, w)$ is never empty.
- ▶ Walrasian demand correspondence is the choice rule induced by the preference relation $\succsim$, i.e.

$$x(p, w) = C_{\succsim}(B_{p,w})$$

Walrasian demand correspondence: Homogeneity of degree zero

Proposition 16

The Walrasian demand correspondence $x(p, w)$ satisfies Homogeneity of Degree Zero: $x(\alpha p, \alpha w) = x(p, w)$ for any p, w and scalar $\alpha > 0$.

- ▶ In words: If prices and wealth change in the same proportion, then the consumer's optimal consumption does not change.
- ▶ Such a change in prices and wealth leaves the budget set and hence the choice problem unchanged. By assumption, for given $\succsim$ the consumer's choice only depends on the budget set and is independent of the framing of the set.

Proof.

For any $\alpha > 0$,

$$B_{\alpha p, \alpha w} = \{x \in X \mid \alpha p x \leq \alpha w\} = B_{p, w} = \{x \in X \mid p x \leq w\}.$$

Thus the UMP is the same and $x(\alpha p, \alpha w) = x(p, w)$.

Walrasian demand correspondence: Walras' Law

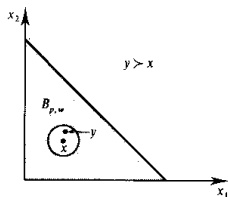
Proposition 17

If $\succsim$ is locally non-satiated, then the Walrasian demand correspondence $x(p, w)$ satisfies Walras' law, that is $px = w$ for all $x \in x(p, w)$.

- The consumer chooses a bundle such that she fully expends her wealth/ on the budget hyperplane.

Proof.

Suppose $px < w$ for some $x \in x(p, w)$. But then there exists another bundle y sufficiently close to x with both $py < w$ and $y \succ x$ (by local non-satiation). But then x is not optimal in the UMP. □



Walrasian Demand Correspondence: Convexity/Uniqueness

Proposition 18

If $\succsim$ is convex/if a utility representation of $\succsim$ is quasi-concave, then $x(p, w)$ is a convex set. If $\succsim$ is strictly convex/if a utility representation of $\succsim$ is strictly quasi-concave, then $x(p, w)$ consists of a single element.

Proof.

Suppose $\succsim$ is convex and that there are two bundles x and x' , such that $x \neq x'$ which are both elements of $x(p, w)$ (i.e. both x and x' maximize $\succsim$ given $B_{p,w}$). Thus $x \sim x'$. By the convexity of $\succsim$, we know that for any convex combination $x'' = \alpha x + (1 - \alpha)x'$ with $\alpha \in [0, 1]$, $x'' \succsim x$ and $x'' \succsim x'$. Moreover, since $B_{p,w}$ is a convex set we know that $x'' \in B_{p,w}$. Then, x'' must also maximize $\succsim$ given $B_{p,w}$ and must be in $x(p, w)$. Thus $x(p, w)$ is a convex set.

Suppose now that $\succsim$ is strictly convex and that there are $x, x' \in x(p, w)$ and $x \neq x'$. Then $x \sim x'$ and $x'' = \alpha x + (1 - \alpha)x'$ with $\alpha \in (0, 1)$ is in $B_{p,w}$ as well (because $B_{p,w}$ is convex). By the strict convexity of $\succsim$, $x'' \succ x$. This however contradicts that $x \in x(p, w)$ and there can be at most one solution to the consumer problem. □

The same proof strategy can be followed using the (strict) quasi-concavity property of u only.

Walrasian Demand Function: Continuity

Proposition 19

If preferences are continuous and strictly convex, then $x(p, w)$ is continuous in prices and wealth.

Proof.

Observe that because of strict convexity and continuity of $\succsim$, $x(p, w)$ is a function. Assume $x(p, w)$ is not continuous. Then, there is a sequence of price and wealth vectors (p^n, w^n) converging to (p^*, w^*) such that $x(p^*, w^*) = x^*$, and $x(p^n, w^n)$ does not converge to x^* .

All prices p_k^n are greater than some positive number p_{min} and all wealth levels w^n are less than some w^{max} . Therefore, all vectors $x(p^n, w^n)$ belong to some compact set (the set of all bundles with no quantity above w^{max}/p_{min}), and thus, without loss of generality (we could choose another infinite subsequence if necessary), we can assume that $x(p^n, w^n) \rightarrow y^*$ for some $y^* \neq x^*$.

Since $p^n x(p^n, w^n) \leq w^n$ for all n , taking limits, we can conclude that $p^* y^* \leq w^*$, that is, $y^* \in B_{p^*, w^*}$. Since x^* is the unique solution for B_{p^*, w^*} , it must be that $x^* \succ y^*$. By the continuity of the preferences, there are balls B_{x^*} and B_{y^*} around x^* and y^* respectively in which the strict preference is preserved. For sufficiently large n , $x(p^n, w^n)$ is in B_{y^*} . Choose a bundle z^* in the ball B_{x^*} so that $p^* z^* < w^*$. For all sufficiently large n , $p^n z^* < w^n$; however, $z^* \succ x(p^n, w^n)$, a contradiction. □